

WICHITA FALLS KICKAPOO, TX, USA

WMO: 720404

Lat: 33.858N

Lon: 98.490W

Elev: 304

StdP: 97.72

Time Zone: -6.00 (NAC)

Period: 07-19

WBAN: 00134

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-7.2	-4.5	-16.2	0.9	-2.6	-13.2	1.2	0.5	11.8	7.9	10.9	6.9	3.6	0	0.525

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.3	40.1	21.5	38.5	22.1	37.3	21.8	25.1	32.9	24.5	32.4	24.0	31.8	4.5	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.0	18.4	27.6	22.6	17.9	27.4	22.1	17.4	27.2	78.1	32.5	75.6	32.4	73.4	31.6	28.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11.0	9.6	8.6	-10.5	41.4	1.9	1.6	-11.8	42.6	-13.0	43.5	-14.0	44.4	-15.4	45.6
			-11.2	25.9	2.2	1.1	-12.8	26.8	-14.0	27.4	-15.3	28.1	-16.8	28.9
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	18.6	6.6	9.1	14.1	18.4	23.0	28.3	29.9	29.7	25.6	18.6	12.7	7.2
	DBStd	9.51	5.30	6.34	5.64	4.67	4.19	2.78	2.64	2.95	3.70	4.91	5.02	5.35
	HDD10.0	394	129	84	26	3	0	0	0	0	0	4	30	118
	HDD18.3	1425	363	263	152	56	12	0	0	0	1	57	177	345
	CDD10.0	3548	25	58	152	254	402	548	616	612	467	271	110	32
	CDD18.3	1538	1	4	19	56	156	298	358	354	218	66	7	1
	CDH23.3	19042	15	61	238	614	1649	3769	4893	4702	2351	650	84	15
	CDH26.7	10228	1	14	63	212	732	2083	2908	2796	1176	224	16	2
Wind	WSAvg	4.1	3.8	4.1	4.7	5.1	4.8	4.6	3.7	3.6	3.7	3.9	3.9	3.6
Precipitation	PrecAvg	763	38	43	58	75	97	88	63	64	67	77	52	42
	PrecMax	1238	80	120	127	214	418	237	185	208	232	169	199	174
	PrecMin	396	1	5	10	9	8	4	0	0	4	0	1	0
	PrecStd	183	24	35	35	53	78	50	48	47	50	46	51	36
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	25.5	28.4	31.3	34.9	37.6	40.3	42.1	42.2	38.0	33.8	28.7	25.9
		MCWB	12.6	14.4	16.3	18.5	20.9	21.6	21.3	21.0	21.3	20.4	17.8	14.7
	2%	DB	21.3	24.9	28.0	31.3	34.9	37.8	40.0	40.2	36.5	31.2	25.7	21.3
		MCWB	11.5	14.4	16.4	17.9	21.4	21.6	21.5	21.4	21.6	19.6	15.6	12.9
	5%	DB	18.7	22.2	26.1	29.0	32.6	36.5	38.0	38.9	34.9	29.1	23.0	18.7
		MCWB	10.2	14.1	15.9	17.5	20.7	22.0	22.2	22.1	21.8	18.8	15.0	12.4
	10%	DB	16.0	19.2	23.6	27.1	30.9	35.1	37.1	37.3	32.9	27.3	21.1	16.0
		MCWB	9.0	12.2	14.7	17.1	20.5	22.6	22.2	22.1	21.4	17.9	14.7	10.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	15.7	18.4	20.2	21.8	24.7	26.0	25.8	25.2	25.1	23.5	20.3	18.0
		MCDB	21.4	23.0	26.7	28.0	32.6	33.7	33.5	32.2	32.7	28.6	24.2	21.5
	2%	WB	13.5	16.9	18.9	20.8	23.7	25.1	25.1	24.5	24.1	22.4	18.2	15.3
		MCDB	18.6	22.3	25.0	27.0	30.6	32.8	33.1	32.6	30.3	27.7	22.0	18.6
	5%	WB	11.3	15.6	17.7	20.1	22.9	24.2	24.5	23.9	23.5	21.2	17.1	13.0
		MCDB	17.2	20.1	22.9	25.9	29.2	31.6	32.7	32.5	29.5	25.8	21.3	17.1
	10%	WB	9.6	13.3	16.5	19.1	22.0	23.6	24.0	23.4	22.9	19.9	15.6	10.6
		MCDB	15.4	18.5	21.5	24.5	28.4	31.1	32.1	31.8	28.9	24.8	20.3	15.6
Mean Daily Temperature Range	5% DB	MDBR	12.9	12.6	13.2	13.1	12.1	12.0	12.3	12.5	11.6	12.5	12.8	11.3
		MCDBR	17.7	17.1	16.3	16.4	15.0	13.8	14.2	14.5	13.7	15.1	15.5	15.9
		MCWBR	9.4	8.7	7.6	7.1	5.2	3.5	3.1	3.3	3.5	6.2	8.1	9.0
	5% WB	MCDBR	15.6	15.6	13.2	12.5	12.1	11.6	11.7	12.2	10.5	11.3	12.4	13.2
		MCWBR	9.4	9.9	6.9	6.7	5.3	4.1	3.4	3.4	3.4	6.3	8.2	9.5
Clear-Sky Solar Irradiance	taub	0.294	0.304	0.338	0.373	0.412	0.427	0.426	0.419	0.387	0.353	0.317	0.298	
	taud	2.520	2.503	2.423	2.351	2.312	2.307	2.348	2.314	2.380	2.461	2.495	2.507	
	Ebn at Noon	926	951	938	914	877	860	857	859	875	882	888	898	
	Edh at Noon	82	93	110	124	130	131	125	127	114	97	84	79	
All-Sky Solar Radiation	RadAvg	3.00	3.59	4.78	5.80	6.22	6.89	6.89	6.31	5.25	4.08	3.17	2.58	
	RadStd	0.31	0.53	0.30	0.47	0.65	0.59	0.43	0.36	0.44	0.59	0.40	0.30	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (42 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page